

ANDREW FISCHER

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Summary

Ambitious, extremely hardworking 4th year electrical and computer engineering student with hands-on experience developing complete embedded systems. I have over 3 years of documented programming experience through work, clubs, and personal projects. In my most recent internship, I developed an annunciator panel used for firefighter training, strengthening skills in robust embedded system design, and long-term maintainability.

Skills

Languages & Protocols: C, C++, Python, MATLAB, Lua, VHDL, git

Embedded Systems: Architectures: AVR, ARM (ARMv6-M), Interfaces: CAN, UART, SPI, I2C, RTOS: FreeRTOS

Design Software: KiCAD, Altium, AutoCAD, LTspice

Lab Equipment: Logic Analyzer, Oscilloscope, Digital Multimeter, Signal Generator

Qualifications: Licensed Amateur Radio Operator - Basic and Advanced (Honours)

Technical Projects

Supercapacitor Bike

- Developed ultra-lightweight ATMEGA328P firmware, including custom I2C, SPI, EEPROM, Timer & PWM drivers using AVR-C and AVR-ASM.
- Designed a dynamic ring buffer to minimize pre-allocated memory and heap corruption.
- Designed custom power electronic circuits to drive a modified BLDC motor, including an efficient N-MOSFET half-bridge circuit to reduce power loss.
- See project [here](#).

Work Experience

Electrical Engineer (Internship)

ContainerWest

JAN. 2025 – AUG. 2025 (8 MO)

Richmond, BC

- Independently designed (KiCAD, C++), tested, documented and presented a working TRL-6 prototype of an expandable annunciator panel capable of monitoring rooms for smoke and fire, allowing firefighters to interact with the panel to simulate events.
- Developed ladder logic programs for OMRON NX1P2 PLCs to automate tasks and monitor live fire burn props to ensure safe operation.
- Gained practical hands-on experience with high-power equipment, including 3-phase AC induction motors, incinomites, and transformers.
- Collaborated with tradespeople and electricians, providing technical guidance and support for installations and troubleshooting.
- Developed ENEFEN-approved system flow graphs and electrical schematics in AutoCAD and performed the associated on-site installations and control system modifications.
- Performed independent site visits and troubleshooting, working directly with clients to ensure reliable and safe operation of the live fire burn props.

Education

University of Victoria

Bachelor of Electrical Engineering, 4th year (CGPA : 3.7/4.0)